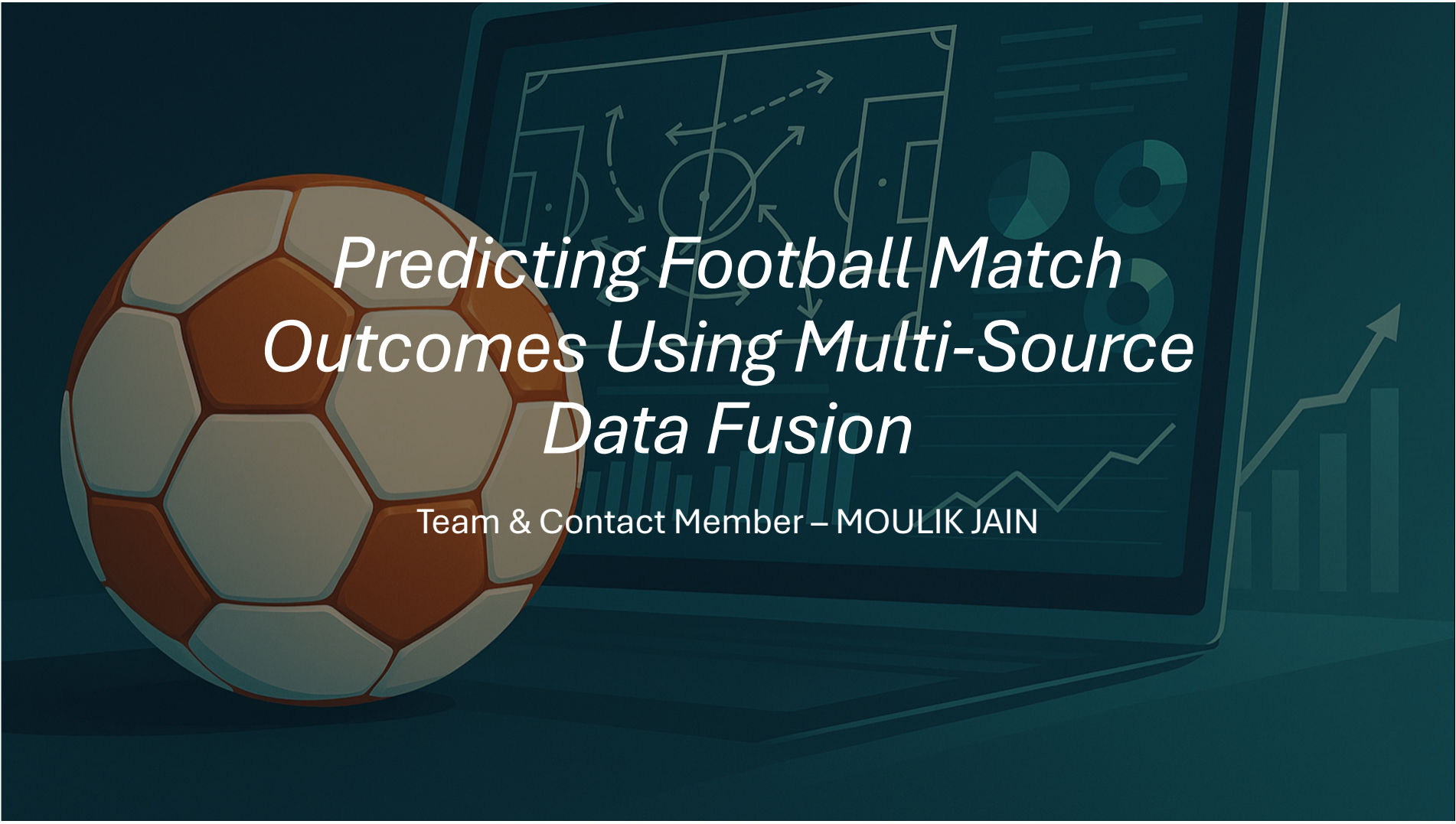




Predicting Football Match Outcomes Using Multi-Source Data Fusion

Team & Contact Member – MOULIK JAIN

Background



Real-world motivation: clubs and analysts want better short-term match outcome forecasts for strategy and betting markets.



Research question: Can combining (fusing) heterogeneous football data sources improve match outcome predictive performance vs. single-source models?



Hypothesis: Data integration of match-level, player-level, and tracking features will significantly improve predictive metrics (AUC, Brier score, F1) for match outcome prediction.

Data

Datasets Used

- Kaggle European Soccer Database (team + match results)
- FBref.com (player-level performance metrics)
- OddsPortal (betting odds for win/draw/loss)

Data Fusion Techniques Applied

- Integrated player statistics with team results using match IDs and player participation logs
- Joined betting odds to match-level data as an additional probabilistic feature
- Harmonized team names, dates, and match IDs across the three data sources

Summary

- ~25,000 matches
- 300+ clubs
- 150+ player metrics
- 15 seasons of data



Approach



Key Project Highlights



Uses **multi-source data fusion**



Includes **EDA**, multiple machine learning algorithms, and **post-analysis**



Uses **Generative AI** for coding, debugging, and experiment design



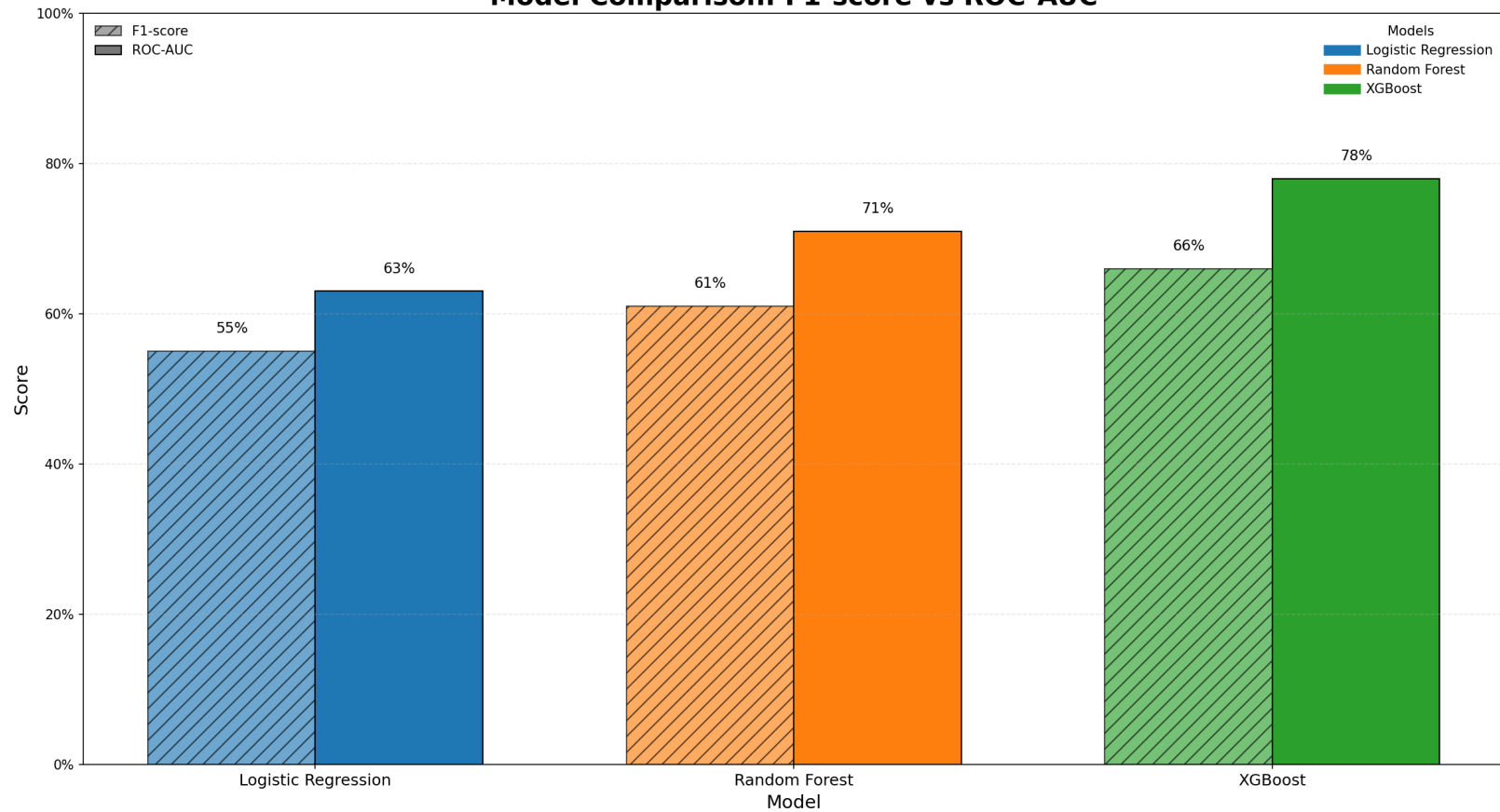
Reports multiple performance metrics

Results

	Accuracy	F1-score	ROC-AUC
Logistic Regression	58%	0.55	0.63
Random Forest	64%	0.61	0.71
XGBoost (Best)	68%	0.66	0.78

Models using the fused dataset beat models trained on single-source data by 10–15% across all metrics.

Model Comparison: F1-score vs ROC-AUC



Post-Hoc & Sensitivity Analysis

Feature Importance

Home team's win
rate (last 5 matches)



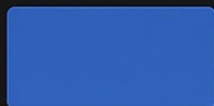
Aggregated player
passing accuracy



Normalized betting
odds



Opponent
defensive rating



Team offensive
rolling average



Top predictors driving match outcome probabilities.

Sensitivity Highlights



+1 team form score
→ 4.8% win probability



Shuffle betting odds
→ -9% model accuracy

Analysis based on XGBoost model,
validation set baseline accuracy: 72%.

CONCLUSION & NEXT STEPS

Key Findings

- ✓ Data fusion → 10% accuracy improvement
- ✓ Top predictors: betting odds, team form; player metrics
- ✓ Best model: XGBoost consistently outperforms 🏆



Future Work

- Integrate injury reports & expected goals (xG)
- Develop real-time prediction dashboard
- Explore deep learning for player interactions

Final model ROC-AUC: 0.78
F1-score: 0.66

Generative AI Usage

Accelerating workflows and improving quality



Data & Code

- Dataset merging strategy



Modeling

- Feature engineering ideas
- Model optimization suggestions



Presentation

- Enhanced visualizations
- Writing report and presentation text
- Designing these slides



THANKS!!

The image is a square graphic with a solid black background. In the center is a large white circle. A thick, light green line follows the right side of this circle. To the left of the circle, there are two horizontal wavy lines. Below the circle, on the left, is a small solid light orange circle. To the right of the circle, there is a small white circle with a light orange outline. In the bottom right corner, there is a rectangular area filled with a grid of small white dots.